



## INSTITUT PENYELIDIKAN PERHUTANAN MALAYSIA

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Rujukan Kami: FRIM(S).600-5/6/12Klt.10(70)

Rujukan Tuan: -

Tarikh: 31 Januari 2022

### SULIT

**NUR FADHILAH MOHD JONI,**

No. 48A, Villa Laman Tasik,  
Bandar Sri Permaisuri,  
56000 Cheras,  
Kuala Lumpur.

Tuan/puan,

### PERKHIDMATAN UJIAN

Dengan segala hormatnya perkara di atas dirujuk.

2. Sukacita dimaklumkan, sebanyak satu (1) sampel tuan/puan (Nama sampel: FFF-Italy; No. makmal: 0301221444) telah diterima oleh pihak kami untuk perkhidmatan ujian.

3. Bersama-sama ini disertakan laporan ujian tersebut untuk perhatian tuan/puan. Laporan ujian ini adalah tertakluk ke atas sampel yang diterima dan bernombor makmal yang dinyatakan sahaja. Pihak FRIM tidak terbabit dengan apa-apa prosiding undang-undang yang berkaitan dengan hasil perkhidmatan.

Sehubungan dengan ini juga, kami sangat menghargai jika tuan/puan dapat memberi maklumbalas untuk penambahbaikan. Sekiranya pihak kami tidak menerima maklumbalas dalam masa dua minggu, maka kami beranggapan anda berpuas hati dengan perkhidmatan yang disediakan.

Sekian, terima kasih.

**"WAWASAN KEMAKMURAN BERSAMA 2030"**  
**"BERKHIDMAT UNTUK NEGARA"**

Saya yang menjalankan amanah,

  
.....  
(MAILINA BINTI JAMIL)

Pegawai Penyelidik  
Bahagian Hasil Semula Jadi  
b/p: Ketua Pengarah FRIM

No Dokumen: **10299945**

**SULIT**



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## CERTIFICATE OF ANALYSIS (COA)

Issue date: 31 January 2022

Name of sample: **FFF-ITALY Agarwood oil**

Sample code/Lab No.: **0301221444 (FFF-ITALY)**

Name of client/company: **NUR FADHILAH MOHD JONI**  
No. 48A, Villa Laman Tasik, Bandar Sri Permaisuri,  
56000 Cheras, Kuala Lumpur.

Date of sample received: **3 January 2022**

Table 1: Chemical compounds in FFF-ITALY Agarwood Oil

| No. | Compound name              | %     |
|-----|----------------------------|-------|
| 1   | 4-phenyl-2-butanone        | 4.65  |
| 2   | 4,5-di-epi-Aristolochene   | 0.29  |
| 3   | $\beta$ -Agarofuran        | 1.79  |
| 4   | Unknown (MW=206.2)         | 0.45  |
| 5   | (E)- $\gamma$ -Bisabolene  | 1.39  |
| 6   | cis-Dihydroagarofuran      | 1.10  |
| 7   | Unknown (MW=218)           | 0.54  |
| 8   | Elemol                     | Trace |
| 9   | $\alpha$ -Agarofuran       | 2.35  |
| 10  | Unknown (MW=194)           | Trace |
| 11  | Unknown (MW=220)           | 0.39  |
| 12  | Unknown (MW=220.2)         | 0.71  |
| 13  | 10-epi- $\gamma$ -Eudesmol | 10.16 |
| 14  | Unknown (MW=204.2)         | 3.16  |
| 15  | $\gamma$ -Eudesmol         | 1.81  |
| 16  | Agarospirol                | 3.06  |
| 17  | Unknown (MW=224.2)         | Trace |
| 18  | Unknown (MW=204.2)         | 9.46  |
| 19  | Valerianol                 | 9.42  |
| 20  | Unknown (MW=218.1)         | 2.08  |

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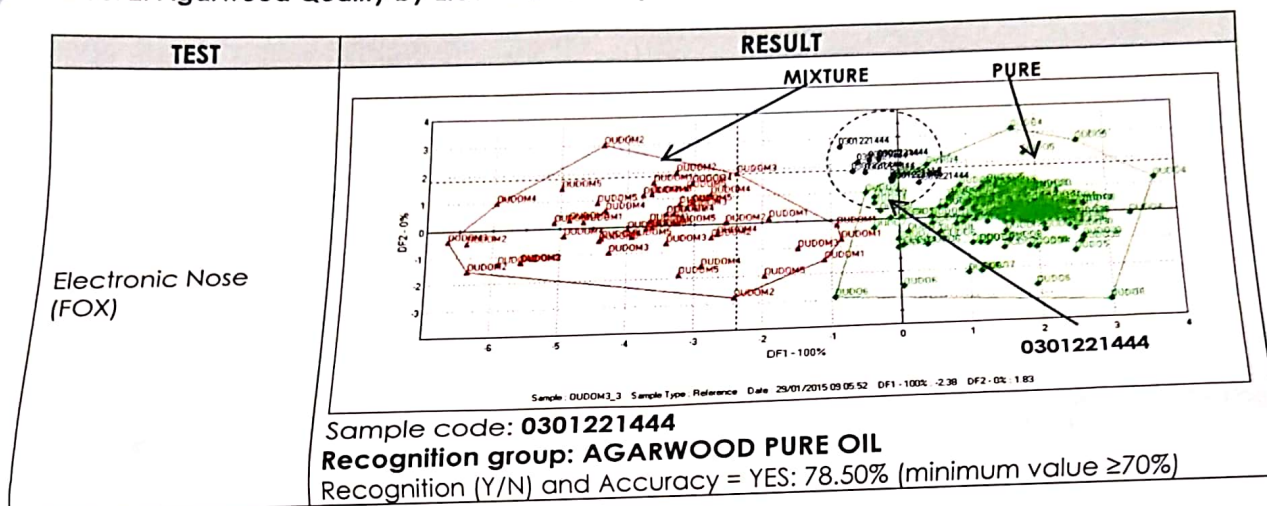
|    |                          |      |
|----|--------------------------|------|
| 21 | Unknown (MW=177)         | 1.80 |
| 22 | Unknown (MW=218.2)       | 1.44 |
| 23 | Unknown (MW=220.2)       | 1.51 |
| 24 | Unknown (MW=218.2)       | 1.00 |
| 25 | Unknown (MW=220.2)       | 1.16 |
| 26 | Unknown (MW=220.2)       | 2.39 |
| 27 | Unknown (MW=220.2)       | 1.09 |
| 28 | Unknown (MW=218.2)       | 2.96 |
| 29 | Unknown (MW=202.2)       | 1.07 |
| 30 | Unknown (MW=189.2)       | 2.43 |
| 31 | Unknown (MW=218.1)       | 0.57 |
| 32 | Unknown (MW=218.2)       | 2.14 |
| 33 | Unknown (MW=218.2)       | 2.61 |
| 34 | epi-Cyclocolorenone      | 1.25 |
| 35 | Dihydrocolumellarin      | 7.37 |
| 36 | 1,5-Diphenyl-3-pentanone | 0.63 |

## GC and GCMS parameter:

| Method                | GC                                        | GCMS                             |
|-----------------------|-------------------------------------------|----------------------------------|
| Model                 | Shimadzu 2010 Plus                        | Agilent Technologies 7890A/5975C |
| Column                | ZB-5                                      | HP-5MS                           |
| Temperature programme | 60°C<br>10min<br>3°C/min<br>230°C<br>1min |                                  |
| Initial temperature:  |                                           |                                  |
| Initial time:         |                                           |                                  |
| Programme time:       |                                           |                                  |
| Final temperature:    |                                           |                                  |
| Hold time:            |                                           |                                  |

**Remark (if any):** The chemical compounds were identified tentatively by matching their mass spectra data with those from the mass spectral database (HPCH2205.L and Wiley7Nist05.L). The relative percentage compositions above were based on the peak areas.

Tabel 2: Agarwood Quality by Electronic Nose (E-Nose)



**Remark (if any):** This test is using FRIM's Agarwood Identification System (AIS) with pre-defined agarwood oils database and classification system with recognition accuracy must be equal or more than 70%. All test results above only valid for the particular sample batch.